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Academic misconduct and artificial intelligence use by medical students, interns and PhD students in Ukraine: a cross-sectional study

Lesya Lymar^{1*} , Iurii Kuchyn¹ , Kateryna Bielka¹ and Livia Puljak^{2*}

Abstract

Background The issues regarding the use of artificial intelligence (AI) and academic integrity are important contemporary topics. There are no clear regulations governing the use of AI in academic institutions in Ukraine. This study aimed to explore the perceptions of medical students, interns, and PhD candidates about academic misconduct and AI use.

Methods The cross-sectional study was conducted online via survey in October–December 2024. The participants were medical students, interns, and medical PhD students of the Bogomolets National Medical University in Kyiv, Ukraine.

Results Among 244 study participants, the majority (84%) reported using AI for academic purposes, with ChatGPT being the most commonly used tool. AI was primarily employed for information searches (70%), while a smaller proportion admitted using it for academic dishonesty, such as writing essays (14%) or submitting pre-written assignments (9%). Nearly half (51%) of participants reported having cheated on tests previously. Opinions on AI's impact on academic integrity were divided, with 36% considering AI use as misconduct, 26% perceiving it as acceptable, and 38% undecided. Most participants viewed AI as beneficial for learning and work, with 37% indicating they would continue using AI professionally. Perceived AI's advantages included time efficiency, enhanced learning, and accessibility, while concerns were raised about errors, lack of critical thinking, over-dependence, and ethical risks.

Conclusion These results highlight the widespread adoption of AI among medical students, interns, and PhD candidates for academic purposes, with both notable advantages and significant ethical and practical concerns. Considering the prevalent use of AI for academic work among medical students, interns and PhD students in Ukraine, national rules are needed to define which uses of AI constitute academic dishonesty and academic misconduct.

Keywords Artificial intelligence, Academic misconduct, PhD students, medical students, interns

*Correspondence:
Lesya Lymar
lesyalymar@nmu.ua
Livia Puljak
livia.puljak@unicath.hr; livia.puljak@gmail.com

¹Bogomolets National Medical University, T. Shevchenko boulevard, 13, Kyiv 01601, Ukraine

²Centre for Evidence-Based Medicine and Health Care, Catholic University of Croatia, Ilica 244, Zagreb 10000, Croatia



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Background

The emergence of artificial intelligence (AI) has significantly influenced various dimensions of education, including medical training. AI-powered applications, such as ChatGPT, have introduced novel opportunities for enhancing learning and providing academic support. However, their adoption has also sparked considerable concern regarding academic integrity and potential misconduct. The ethical implications of AI use in academic environments—particularly within medical education—remain the subject of ongoing global debate. In Ukraine, the relevance of these concerns intensified in February 2023, when the Ministry of Digital Transformation officially enabled access to ChatGPT for Ukrainian users [1]. While the tool was welcomed for its potential to support educational activities, it was also met with apprehension due to its possible misuse in academic settings.

To date, no official recommendations or prohibitions have been issued regarding the use of AI in academic activities in Ukraine. A so-called “White Paper” published by the Ministry of Digital Transformation offers general recommendations but does not establish binding rules or regulations [2]. The absence of clear guidelines on academic integrity in relation to AI use is particularly significant in the context of medical education, where academic honesty is a foundational professional value.

Academic integrity is widely regarded as an essential characteristic of students; however, incidents of cheating and plagiarism are well-documented across higher education institutions [3]. Previous studies by Chung et al., published in 2019 [4] and Đogaš et al., from 2014 [5] have demonstrated that academic dishonesty is prevalent among medical students. In these studies, cheating was defined as presenting another person's work as one's own.

This raises a contemporary ethical dilemma on whether the use of AI constitutes academic misconduct, since AI is a tool rather than a person. However, depending on how it is used, particularly if it replaces critical thinking or misrepresents authorship, the use of AI may lead to unethical academic behavior.

Scholars addressing the intersection of artificial intelligence (AI) and academic integrity express divergent views on whether the use of AI constitutes academic misconduct. In 2023, Anders argued that students may exploit semantic ambiguities to justify the use of AI for writing test papers or essays, claiming that such use does not amount to cheating because AI is a tool, not a person [6].

In 2024, Kazley et al. published results indicating that medical students often do not perceive AI use as a form of cheating [7]. Their study revealed that students regarded AI assistance as beneficial and necessary, without associating it with signs of misconduct [7].

Medical education is distinct from other educational domains due to the heightened ethical and professional responsibilities assumed by its graduates. Academic dishonesty in medical training has serious implications, as it may compromise future clinical competence and potentially endanger patient lives. Consequently, concerns about the ethical use of AI are especially pronounced in this context.

In 2023, Mohammad et al. published a scoping review on the use of ChatGPT in medical education, concluding that it can be a valuable tool, particularly for both educators and students, by offering diverse informational resources and problem-solving support [8]. Xu et al. (2024) similarly emphasized that ChatGPT facilitates the management of large volumes of information, a task often overwhelming for medical students [9]. This perspective was also supported by Boscardin et al. [10] and Preiksaitis and Rose [11], who highlighted the tool's potential to enhance learning and support decision-making in medical training.

Most medical students recognize cheating as a violation of academic integrity and acknowledge it as a breach of institutional rules. This recognition raises an important question about whether the use of AI, such as ChatGPT, similarly constitutes a violation of academic integrity. Liu et al. [12] and Miao et al. [13] have explored the boundaries of appropriate AI use in medical academic writing. While both studies acknowledge the educational benefits of AI tools, they also underscore the ethical risks associated with their misuse, particularly in contexts that demand originality and authorship [12, 13].

Despite the growing integration of AI tools into medical education, the boundary between legitimate educational use and academic misconduct remains ill-defined. Specifically, it is unclear at what point the use of ChatGPT, or similar AI systems, transitions from being a study aid to constituting a form of academic dishonesty. Furthermore, there is a notable lack of empirical research examining how medical students perceive the educational use of AI and whether they view AI-generated content as ethically problematic or equal to cheating. Gaining a deeper understanding of these perceptions is essential, especially in light of the high ethical and professional standards expected of future medical practitioners.

While the potential of AI to support learning is widely acknowledged, its use may also inadvertently lead to breaches of academic integrity if users are unaware of its limitations. Accordingly, there is an urgent need for the academic community to establish clear guidelines and policies that delineate acceptable and unacceptable uses of AI in educational settings. As of December 2024, Ukraine has not introduced national regulations specifically addressing the ethical use of AI in academic contexts.

This study aimed to explore medical students' interns, and PhD candidates' perceptions of AI use in academic settings, their views on its ethical implications, and whether they associate AI-assisted work with academic misconduct. Findings from this study can inform policy development and contribute to future guidelines for the responsible use of AI in medical education in Ukraine and beyond.

Methods

Study design

A cross-sectional study was conducted using a convenience sample.

Setting

The study was conducted between October and December 2024 at the Bogomolets National Medical University (NMU), Kyiv, Ukraine.

Participants

Eligible participants were medical students, interns, and medical PhD students enrolled at the Bogomolets NMU, Ukraine, at the time of the study. Participants were excluded if they did not belong to the above academic categories (e.g., other health professions students, faculty, administrative staff), declined to participate, or did not provide informed consent. All individuals in the eligible target population were invited to participate.

Ethics

The Bogomolets National Medical University has a policy that the cross-sectional studies on healthy participants are exempt from the need to obtain approval of the ethics committee, i.e., such studies are eligible for a waiver. The Ethics Committee of the Bogomolets National Medical University waived the need for an ethics appraisal of the study protocol, as it was a cross-sectional study on healthy volunteers (official document to confirm the waiver was issued with a decision number 120/3 from December 27, 2024).

The participants voluntarily participated in the survey upon being sent an invitation to participate, along with detailed information about the study, through Google Classroom. All participants provided written informed consent in the Google Form before starting to participate in the survey. Participants were allowed to withdraw from or stop the survey at any point before submitting it. The Google Form contained no previous information that could affect the answers. The study was carried out in compliance with the Helsinki declaration, and in line with all the applicable laws and regulations. To prevent unauthorized access, the first author (LL) stored the data on a password-protected server.

Data collection

The participants were asked to participate in the study through the mobile app communication channels within the academic groups. The initial contact with the participants was made through the academic messenger groups. The participation was voluntary and anonymous. The authors offered no incentives to the invited individuals.

Questionnaire

The participants completed a 17-item questionnaire developed specifically for this study. The questionnaire was developed and administered in Ukrainian. A new questionnaire was designed, as no previously validated instrument in Ukrainian was identified that addressed both academic misconduct and the use of AI in academic settings among medical students, interns, and PhD candidates.

The questionnaire was created following in-depth discussions among Bogomolets NMU faculty regarding observed cases of academic misconduct and patterns of AI use, to assess both learning-related and unethical uses of AI. Drawing on published literature on AI in medical education and academic integrity, and on institutional discussions of observed behaviors, we specified key content domains of the questionnaire (AI awareness and access; patterns of academic use; ethically problematic use; prior misconduct; perceived usefulness/risks; open-ended advantages/disadvantages). Draft items were written in Ukrainian with plain, unambiguous wording and fixed response options where appropriate.

A small panel of Bogomolets NMU faculty with expertise in medical education, ethics/academic integrity, and assessment conducted an independent, parallel review of the draft instrument. Each reviewer examined the item stems, response options, instructions, and section flow against four a priori criteria, including relevance to the study objectives, coverage of the intended content domains, clarity of wording, and alignment with the target learner population. Reviewers annotated the draft using tracked changes and margin notes, indicating an action for each item (retain as is/revise/consider removal). Comments were then synthesized in a brief adjudication meeting to resolve discrepancies and agree on edits. Revisions were minor and non-substantive but improved usability. We simplified several stems to avoid double-barreled phrasing and standardized tense and person across items. We also checked for neutral tone in items addressing academically sensitive behaviors and ensured consistent ordering and grouping of questions to support respondent flow.

For cognitive pre-testing, the draft instrument was piloted among study authors to evaluate comprehension, interpretation, and completion experience. We

noted points for misinterpretation, assessed whether answer categories were mutually exclusive and collectively exhaustive, and recorded approximate completion time under typical conditions. We also verified on-screen instructions, item order, and confirmed functionality on both desktop and mobile. Feedback led to minor refinements, including shortening long stems and adjusting a few response labels for precision.

Minor wording adjustments were made based on their feedback. The final instrument is presented in Appendix 1. For the writing of this manuscript, the questions and the answers to open-ended questions were translated from Ukrainian into English.

For this study, we defined AI use as the use of explicit, user-invoked generative AI tools (e.g., ChatGPT/GPT chat, Gemini, Bing chat). We did not separately assess implicit AI features embedded within productivity software, including Microsoft Word and PowerPoint, among others.

The questionnaire included the gender, age, and educational level of participants, as well as items regarding the participants' attitudes about AI use in their studies and their perceptions of cheating and academic integrity. The participants were asked whether they use AI in their studies, how they use it, how they learned about AI opportunities, the frequency of AI use, their self-assessment of the difficulty of work and study without AI, the ways they use AI, and whether they use AI for essay writing. They were also asked if they had ever cheated by copying test answers from a classmate, downloading essays from the Web and submitting them as their own, or buying essays. There were questions about their perception of academic integrity in using AI for writing essays, making tests, taking exams, etc. The participants were also asked to state the level of usefulness of the AI and assess the probability of using the AI in their future work.

The last two questions were open-ended; the participants were asked to list the advantages and disadvantages of using AI in the studies. Personal data were not collected or stored in the survey. The participants were not informed about the survey's results upon its completion. The cookies were not used. IP check was not used. Log file analysis was not used. The registration was based on the participant's Google account. No questionnaires were left incomplete, as an answer was required for each question. The completion rate was 100%. No questionnaires were filled in with an atypical timestamp. No weighting of items was used.

Adaptive questioning was not used. The participants could change their answers until they pressed the final "submit" button. All those who used the link filled out the form.

Data analysis

Descriptive data analysis was used; the data were presented as frequencies and percentages. Data were

analyzed using MedCalc software (MedCalc Software Ltd., Ostend, Belgium).

Open-ended questions were analyzed using the qualitative description (QD) method [14]. QD is particularly suitable for analyzing textual data in applied research contexts. This approach facilitates the straightforward interpretation of responses without imposing theoretical frameworks or excessive abstraction. First, the participants' responses were translated into English and then analyzed. Two researchers (LL and LP) independently read all responses multiple times to become familiar with the data and gain an initial understanding of emerging patterns. Using an inductive thematic approach, one researcher (LP) coded responses line by line, identifying key phrases and concepts that captured participants' perspectives. A data-driven coding framework was developed iteratively, allowing new themes to emerge naturally from the data rather than being pre-determined. Another researcher (LL) verified the approach independently. To resolve coding disagreements, we planned to use a structured consensus workflow to handle discrepancies and to include a discussion phase. However, we did not have any disagreements in the independent verification.

The open-ended responses were grouped into key themes and sub-themes based on their content and recurring patterns. Representative quotes were used to illustrate findings while maintaining the integrity of participants' original responses. The participants' quotes were labeled with alphanumerical codes denoting a participant (P) and the consecutive number of responses (such as P1). Reflexivity and credibility were maintained through researcher triangulation, iterative discussions, and an audit trail of coding decisions. This method ensured that the findings remained faithful to participants' experiences, providing practical insights grounded in real-world perspectives while minimizing researcher bias. We did not use any software for the analysis of qualitative data.

Reporting

The study was reported in line with the CHERRIES checklist (Eysenbach, 2004).

Results

Among 5000 eligible individuals (3500 medical students, 800 interns, 700 PhD students), 244 individuals participated in the study, including 183 medical students (75%), 6 interns (2%), and 55 PhD students (23%). A flowchart of participant selection and inclusion in the study is available in Appendix 2. Most of the participants were under 25 years old (78%) and women (69%) (Table 1).

Most participants stated they used the AI for their study (for any purpose). A small portion denied using AI for academic purposes. The most widely used AI was

Table 1 The use of AI among medical students, interns and PhD candidates for their academic work

Study questions	Medical students (N=183)	Interns (N=6)	PhD students (N= 55)	All participants (N=244)
Age:				
17-25	182 (99.5)	0	8 (15)	190 (78)
26-35	1 (0.5)	6 (100)	28 (51)	35 (14)
36-45	0	0	14 (25)	14 (5.7)
Over 45	0	0	5 (9.0)	5 (2.0)
Gender:				
Man	55 (30)	2 (33)	16 (29)	73 (30)
Woman	127 (69)	4 (67)	38 (69)	169 (70)
Other	1 (0.5)	0	1 (1.8)	2 (0.8)
Do you use the AI in your studies?				
Yes	153 (86)	5 (83)	46 (84)	204 (84)
No	30 (14)	1 (17)	9 (16)	40 (16)
What AI tool do you use:				
GPT	121 (66)	4 (69)	36 (65)	161 (66)
Gemini	20 (11)	1 (16)	6 (10)	27 (11.0)
Bing	3 (1.6)	0	1 (1.8)	4 (1.6)
Other	10 (5.4)	0	3 (5.4)	13 (5.3)
Don't use	29 (15)	1 (16)	9 (17)	39 (16)
How did you start using the AI?				
I don't remember	48 (26)	2 (40)	14 (25)	64 (26)
I found it on the web, I googled it	43 (24)	1 (20)	13 (24)	57 (23.0)
My friends/relatives told me about it	83 (45)	3 (60)	25 (45)	111 (45)
My teacher told me about it.	9 (4.9)	0	3 (5.4)	12 (4.9)
How often do you use AI for your study?				
I don't use it for study	32 (17)	1 (20)	10 (18)	43 (18)
Once a week	73 (40)	2 (40)	22 (40)	97 (40)
2-3 times a week	43 (23)	1 (20)	13 (22)	57 (23)
More than 3 times a week	35 (20)	1 (20)	11 (20)	47 (19)
If you are not be able to access AI any more, how hard will it be for you to study/research?				
Very difficult	7 (3.8)	0	2 (3.6)	9 (3.6)
Difficult	10 (5.4)	0	3 (5.4)	13 (5.3)
Undecided	22 (12)	1 (17)	7 (13)	29 (12)
Not quite difficult	58 (32)	2 (33)	18 (33)	78 (32)
Not difficult	86 (47)	3 (50)	26 (46)	115 (47)
If you use the AI for study in the University, you:				
Search information	128 (70)	4 (67)	38 (70)	170 (70)
Use the information from the AI as mine work	8 (4.3)	0	2 (3.6)	10 (4.1)
Use the information from the AI, but adapt it, paraphrase	48 (26)	2 (33)	14 (26)	64 (26)

ChatGPT, followed by Gemini. The participants mostly started using AI on the recommendation of friends. Most of them used the AI once or more than once a week for their study. If the AI disappears, most indicated they would experience no difficulty (Table 1).

Table 2 Using AI for cheating and academic misconduct

Study questions	Medical students (N= 183)	Interns (N=6)	PhD students (N=55)	All participants (N=244)
Did you ask the AI to write essays for you?				
Yes	25 (14)	1 (17)	7 (13)	33 (14)
No	158 (86)	5 (84)	48 (87)	211 (87)
Have you submitted the essays downloaded from the web as your own?				
Yes	16 (8.7)	1 (17)	5 (9.0)	22 (9.0)
No	166 (91)	5 (84)	50 (91)	222 (91)
Have you used the services of companies that offer to write the essay for you?				
Yes	46 (25)	2 (33)	14 (25)	61 (25)
No	137 (75)	4 (67)	41 (75)	183 (75)
Perception of the AI use for doing tests, writing essays, and coursework:				
Academically righteous	48 (26)	2 (33)	14 (25)	64 (26)
Academically wrong	65 (36)	2 (33)	20 (36)	87 (35)
Undecided	70 (38)	2 (33)	21 (39)	93 (38)
Usefulness of the AI for the study:				
Useful	64 (35)	3 (50)	19 (35)	85 (35)
More useful than useless	74 (40)	2 (33)	22 (40)	98 (40)
Undecided	29 (16)	1 (17)	9 (16)	39 (16)
More useless than useful	13 (7.1)	0	4 (7.2)	17 (6.9)
Useless	4 (2.1)	0	1 (0.5)	5 (2.0)
Have you ever cheated by copying the test answers from a classmate?				
Yes	93 (51)	3 (50)	28 (51)	124 (51)
No	74 (40)	2 (33)	22 (40)	99 (41)
I can't say	16 (8.7)	1 (17)	5 (9.0)	21 (8.6)
Will you use the AI results at work in future:				
Yes	68 (37)	3 (50)	20 (36)	90 (37)
No	40 (22)	1 (33)	12 (22)	54 (22)
It's hard to say	75 (41)	2 (17)	23 (42)	100 (41)

Most participants used AI for information search, and only a few admitted to using AI for direct cheating or writing essays. The same proportion confessed to downloading the essays from the Web or having bought the essays for money. Almost half of the participants had cheated at tests (Table 2).

As for their perception of AI and academic integrity, 26% considered it does not violate the academic integrity rules, 36% indicated that AI use is academic misconduct, and 38% still could not define. Most of them considered the AI to be useful for work and studies, and one-third will continue using it for work (Table 2).

Participants' perceptions of the advantages of AI use

There were 232/244 (95%) participants who answered the question about the advantages of AI use. The six key themes identified were: time efficiency, simplification of complex information, enhanced learning and understanding, accessibility of information, accuracy, and precision. The key themes, sub-themes, and representative quotes are shown in Table 3.

Table 3 Themes, sub-themes and representative quotes from the participants' open-ended responses about the advantages of AI use

Themes	Sub-themes	Representative participants' quotes
Time efficiency	Quick access to information	"AI provides quick information" (P15) "AI brings different types of information in seconds." (P32)
	Faster task completion	"AI speeds up doing tasks" (P126) "It reduces the time spent on research." (P102)
	Time economy	"It saves time as you get all answers at once" (P138) "It can save your time doing monotonous work." (P134)
Simplification of complex information	Breaking down difficult concepts	"AI can simplify difficult information" (P7) "AI simplifies technical information for students" (P74)
	Data synthesis	"It transforms raw data into analyzed information" (P225) "It enhances the ability to synthesize information." (P69)
Enhanced learning and understanding	Personalized learning	"AI adapts to specific needs." (P72) "It personalizes the search for PhD-level queries." (P92)
	Improved understanding	"A source for understanding complex topics." (P10) "It explains unclear concepts." (P18)
Accessibility of information	Ease of access	"AI provides access to any information" (P135) "It gives access to hidden sources." (P90)
	Relevance	"AI eliminates irrelevant information" (P29) "AI offers content for specific issues." (P54)
Automation and task delegation	Homework and assignments	"AI completes homework tasks, both for me and my child at school." (P171) "AI will do any homework you need" (P211)
	Data processing	"AI organizes data for effective literature reviews" (P93) "It processes vast amounts of information quickly." (P128)
Accuracy and precision	Accurate data handling	"AI ensures accuracy." (P62) "It provides updated and accurate information." (P84)
	Reliable outputs	"AI is more useful than encyclopedias" (P39) "Reliable access to niche knowledge." (P89)

The most frequently mentioned theme was time efficiency, with over 50% of participants highlighting AI's ability to save time and streamline processes. Relevant sub-themes regarding time efficiency were quick access to information, faster task completion, and time economy. Simplification of complex information and accessibility of information were the themes most frequently mentioned, indicating AI's perceived educational value. Relevant sub-themes highlighted by the participants were the AI's ability to break down complex concepts and its use for data synthesis (Table 3).

Participants highlighted AI's role in education and research, aiding comprehension and providing tailored assistance. Sub-themes identified for enhanced learning and understanding were personalized learning and improved understanding. AI's ability to provide a wide range of accessible and relevant information was noted as a key advantage, with sub-themes highlighting ease of access and relevance (Table 3).

Automation and task delegation were noted as another advantage by the participants, with sub-themes regarding AI's importance for doing homework and assignments and data processing. AI was praised for its accuracy and precision, with sub-themes of accurate data handling and reliable outputs (Table 3).

Participants' perceptions of the disadvantages of AI use

Among 208/244 (85%) of participants who responded to the question about the disadvantages of AI use, the following seven key themes were identified: errors and inaccuracy, lack of critical thinking and contextual understanding, dependency and skill atrophy, ethical concerns, lack of responsibility for mistakes, overuse and dependency, and futuristic and philosophical fears. The key themes, sub-themes, and representative quotes are shown in Table 4.

The most frequently mentioned disadvantage of AI by more than 60% of the participants was its tendency to provide incorrect or incomplete information, undermining its reliability. Sub-themes identified for this theme were mistakes and inaccuracies, superficial outputs, and inability to verify or explain errors (Table 4).

Participants highlighted AI's inability due to a lack of critical thinking and contextual understanding, with sub-themes of contextual misinterpretation, lack of logic, and inability to evaluate critically. Over-reliance on AI was perceived as a threat to developing critical thinking and analytical skills. This dependency and skill atrophy theme included the following sub-themes: loss of analytical abilities, erosion of practical skills, and over-dependence (Table 4).

Table 4 Themes, sub-themes and representative quotes from the participants' open-ended responses about the disadvantages of AI use

Themes	Sub-themes	Representative participants' quotes
Errors and inaccuracy	Mistakes and inaccuracies	"AI sometimes provides incorrect answers" (P1) "It can misinterpret questions" (P9) "Many mistakes." (P4) (P6) (P18) (P61) (P66) (P86) (P105) (P113) (P138)
	Superficial outputs	"AI offers superficial information" (P10) "It overlooks basic facts." (P29)
	Inability to verify or explain errors	"AI cannot explain why an error occurred" (P30) "No accountability for errors." (P30) (P99)
Lack of critical thinking and contextual understanding	Contextual misinterpretation	"AI does not understand context" (P21) "It confuses terms." (P22)
	Lack of logic	"No logical reasoning." (P133) "Primitive logics and mental abilities." (P97)
	Inability to critically evaluate	"AI cannot critically evaluate anything." (P101) "Its responses lack critical review." (P62)
Dependency and skill atrophy	Loss of analytical abilities	"We will become stupid if we rely only on AI" (P142) (P149) "It reduces mental activity." (P147)
	Erosion of practical skills	"Students may forget how to solve problems on their own" (P154) "It takes away experience of data search and analysis." (P155)
	Over-dependence	"We can't solve a problem if the AI is absent" (P155) "AI replaces logic." (P157)
Ethical concerns	Facilitation of cheating	"AI makes copying easier" (P157) "It encourages submitting work that's not your own." (P169)
	Compromise of originality	"AI can compromise originality" (P176) "It increases risks of academic dishonesty." (P174)
Lack of responsibility for mistakes	No accountability	"AI cannot take responsibility for mistakes" (P2) "Who is responsible for the mistakes?" (P60)
	Excuses for errors	"When you find a mistake and show it, it just says sorry" (P5) "Many mistakes, and excuses when you show them." (P46)
Overuse and dependency on AI	Overuse diminishes cognitive ability and creativity	"Overusing AI dulls creativity" (P158) "AI reduces mental activity." (P147)
	Long-term effects	"In a generation, we will lose thinking skills" (P163) "Machines may substitute people as they will not be able to think." (P153)
Futuristic and philosophical fears		"It's scary, in Terminator it all also started with machines. AI can capture us." (P208)

It was noted that AI's ease of use can lead to ethical concerns related to academic dishonesty and plagiarism. Sub-themes related to ethical issues were the facilitation of cheating and the compromise of originality. Furthermore, a recurrent theme was the lack of AI's responsibility for mistakes, with sub-themes of no accountability and excuses for errors.

Participants saw overuse and dependency on AI as potentially harmful to human creativity, learning, and problem-solving skills. Sub-themes identified for this theme were concerns that AI overuse diminishes cognitive ability and creativity, and its long-term effects on thinking skills.

The final theme included futuristic and philosophical fears (Table 4). These fears included loss of human agency (e.g., fears that "machines will take over" or humans will become subordinate to automated systems), erosion of the human elements of medicine (e.g., concerns that AI could displace judgment, empathy, or the clinician–patient relationship), and dystopian framings drawing on

popular culture (e.g., Terminator-style scenarios). While this theme was less frequently mentioned than others, we report it for completeness because it reflects how some learners contextualize near-term educational concerns within longer-term societal narratives. We include additional illustrative quotations in Table 4 alongside the existing example.

Of note, 13% (31/244) of participants noted no disadvantages associated with AI use. On a humorous note, one of the participants stated, "*No machine has suffered during filling out of this form*".

Discussion

This study revealed extensive use of AI among medical students, interns, and PhD candidates, with ChatGPT being the most commonly utilized tool. While 84% of participants used AI for information-seeking and academic support, a notable minority admitted to using it for dishonest practices such as essay writing and test cheating. Perceptions of AI's alignment with academic

integrity were divided, with over one-third categorizing AI use as misconduct. These findings reflect both the growing reliance on AI in academic settings and the ethical uncertainty surrounding its use in medical education.

It is of particular concern that almost half of the participants indicated they have violated academic integrity before and confessed to using AI for academic misconduct. This requires a clear definition of the limits of AI in the study and research, particularly in Ukraine, where no definite regulations on AI use exist in education and research. In 2023, the United Nations Educational, Scientific, and Cultural Organization (UNESCO) published the first-ever global Guidance on Generative AI in Education and Research, which was designed to address the disruptions caused by generative AI technologies [15].

Simultaneously, UNESCO called on governments to implement appropriate regulations and teacher training to ensure a human-centered approach to using generative AI in education [16]. As for Ukraine, the “White Paper” on the AI implementation for academic activity has been issued, which contains recommendations, but it does not include strict rules [2]. The Ministry of Education and Science of Ukraine has issued the Project for Instructions and Methodical Recommendations on AI use in Educational Establishments, which contains general phrases about “*the necessity of keeping to the academic integrity rules while using the AI*” (Ministry of Digital Transformation of Ukraine, 2024).

Several governmental bodies worldwide have issued guidance on using generative AI in education. In the United States, in October 2023, following a landmark Executive Order by President Biden, the U.S. Department of Education committed to developing resources, policies, and guidance focused on the safe, responsible, and accessible use of AI in education [17]. In the UK, the governmental Department of Education has recognized the potential of generative AI to assist educators by reducing administrative burdens and providing tailored support. They have published guidance discussing the benefits and emerging evidence on the risks associated with AI use in educational settings [18].

Answers regarding the advantages of AI use in academic work revealed that participants viewed AI primarily as a tool for improving efficiency, simplifying processes, and enhancing learning. They indicated it bridges gaps in understanding, accelerates workflows, and provides reliable, easily accessible information. These findings highlight AI’s transformative potential in academia, research, and personal productivity. Other studies support these statements. Dave et al., in their overview, claimed that AI could be widely used in medical research for planning the process, analysis of the literature, and clinical analysis data treatment [19]. Coraci et al. tested the usefulness of ChatGPT for assessing low back pain.

They concluded that ChatGPT could help evaluate patients, including multilevel perspectives. However, they highlighted that the power of ChatGPT is limited and recommended further research and validation [20].

Several recent reviews confirm the widespread adoption of AI tools in medical education and research, emphasizing both their pedagogical benefits and the risks to academic integrity if misused [8, 9, 21–25].

Our findings echo those of Mohammad et al. [8] and Xu et al. [9], who identified AI’s utility in enhancing medical students’ access to knowledge and reducing cognitive load. However, concerns expressed by our participants, such as overreliance, misinformation, and ethical ambiguity, align with warnings raised by other studies [24–26] about the limitations and potential misuse of AI in academic settings. These perspectives resonate with our participants’ own insights, who simultaneously valued AI’s efficiency and expressed concerns over its reliability, ethical implications, and cognitive impacts.

Cooper and Rodman argued that AI, particularly tools like ChatGPT, was rapidly transforming medical education, presenting both opportunities and challenges [27]. While AI demonstrated value in supporting clinical reasoning, learning, and curriculum development, it also raised concerns about misinformation (“hallucinations”), ethical misuse, and the potential erosion of traditional cognitive apprenticeship models. The authors emphasized that medical educators could no longer afford to take a passive stance; instead, they needed to actively shape how AI was integrated into training. They called for the development of adaptive curricula, careful assessment of AI applications, and clear regulatory guidance to ensure that AI would enhance, rather than compromise, medical professionalism and academic standards [27].

Recent literature mirrors our findings of high AI adoption with mixed views on integrity and learning outcomes. Large surveys of medical students published in 2025 reported widespread use of ChatGPT for study support, with perceived benefits for explanation and drafting, but persistent concerns about misinformation and academic honesty – patterns consistent with our cohort’s views [28, 29]. Those surveys found high awareness and regular use among medical undergraduates, alongside divided attitudes about whether use constitutes misconduct and worries about over-reliance and accuracy [28, 29].

Evidence synthesis of the use of AI in medical education also underscores a dual reality of a pedagogical promise paired with integrity and assessment challenges. A scoping review of 23 studies published in 2025 showed that generative AI can support knowledge access and efficiency, yet raises risks around plagiarism, authorship ambiguity, and erosion of critical thinking if unstructured or unassessed [30].

Another scoping review, published in 2024, argued for structured AI literacy rather than prohibition. Revision of medical-AI curricula and competency frameworks should incorporate explicit training in evidence-based use, limitations (e.g., hallucinations), citation/attribution, and assessment redesign, while aligning competencies with clinical decision-making [31].

Global policy guidance (e.g., UNESCO's updated "Guidance for generative AI in education and research") likewise supports institution- and ministry-level policies that emphasize transparency, disclosure, and ethics training [32].

Taken together, this comparative evidence strengthens our findings that AI use is already pervasive among health-professions learners and perceptions of academic integrity are ambivalent. Thus, the most promising path forward is not restriction but the integration of AI literacy and ethics into curricula, paired with assessment practices that reward critical engagement over unacknowledged AI assistance.

International evidence suggests our findings mirror broader trends. Student uptake of generative AI for learning is now near-universal, including for assessed work, while norms about "acceptable" versus "misuse" remain ambiguous. A study conducted in December 2024 found that 92% of UK undergraduates now use genAI, and 88% report using it for assessments [2]. At the same time, studies indicate AI may be shifting how academic dishonesty occurs rather than clearly increasing overall self-reported cheating, with high-school survey data showing no post-ChatGPT jump in cheating prevalence [33]. Nonetheless, confirmed AI-related misconduct cases recorded by universities are rising and institutions are stress-testing assessment to respond [34]. Against this backdrop, and given Ukraine's 15 May 2025 signature of the Council of Europe Framework Convention on AI and the EU AI Act's designation of several education uses (e.g., admissions, evaluating learning outcomes, remote proctoring) as high-risk with governance and human-oversight duties, policy should move beyond general guidance to enforceable standards [35].

Policy implications for Ukraine

Ukraine currently lacks binding national rules on educational AI use; existing documents offer recommendations rather than enforceable standards. Thus, we propose implementing a pragmatic, staged policy package. A government-issued document could define acceptable AI-supported tasks (e.g., language polishing, idea generation) vs. misconduct (ghost-writing, undisclosed AI authorship), and require student disclosure of substantive AI assistance in coursework. Educational institutions could require mandatory course-level AI statements, and prioritize in-class, oral, or process-evidence assessments

(drafts, rationales) to reduce opportunities for substitution. Regarding AI literacy and integrity training, institutions should integrate short, required modules for students and faculty on evidence-based use, citation/attribution, hallucinations, and data/privacy. Institutional codes should be updated to align academic integrity codes to explicitly cover undisclosed AI use as a form of misrepresentation; provide proportionate sanctioning guidance. Tool governance implies maintaining an institutional list of permitted tools and data-handling rules (no sensitive/patient data; logging off personal accounts). Detection safeguards should discourage sole reliance on AI-detectors for adjudication and require triangulated evidence (process logs, drafts, oral defense). For capacity building, there should be funds for faculty development, exemplar assignments, and repositories of AI-resilient assessment tasks. Finally, monitoring would be beneficial to collect annual statistics on cases, student literacy, and outcomes to refine policy. These steps are consistent with UNESCO's human-centered approach and sector principles adopted across leading university groups [15].

Limitations

This study had several limitations. The overall response rate was 4.9% (244/5,000), with 5.2% among students, 0.8% among interns, and 7.9% among PhD candidates. The achieved sample over-represented PhD candidates (23% vs. 14%) and under-represented interns (2% vs. 16%) relative to the target frame, raising the possibility of non-response/selection bias if AI use or integrity attitudes vary by academic level. We therefore caution against generalizing point estimates to all eligible learners at the university and emphasize subgroup-specific results. Similar trends have been observed in other research conducted in conflict-affected settings, where academic engagement can be deprioritized due to pressing personal and societal concerns [36].

The study relied on a convenience sample from a single university, limiting the generalizability of findings to other institutions or broader populations. A self-report bias cannot be excluded. The survey did not explore the depth of AI usage, such as variations in AI-assisted learning strategies or the impact on knowledge retention and critical thinking skills. The cross-sectional nature of the study captures perceptions at a single time point, preventing the assessment of changes in AI adoption, attitudes, and academic integrity over time. The study focused solely on student perspectives, without input from faculty or administrators, limiting the broader understanding of AI's role in medical education.

In this study, we used a newly developed questionnaire. Because items were predominantly single indicators intended to capture distinct behaviors and perceptions (formative content), we did not compute

internal consistency (e.g., Cronbach's α) or perform factor analysis. Instead, we relied on expert review and cognitive pre-testing to support content and face validity.

Future directions

Future research should explore how attitudes toward AI and academic integrity evolve over time and across diverse educational settings. Expanding the study to multiple universities and incorporating qualitative methods, such as interviews or focus groups, could provide deeper insights into students' reasoning and ethical considerations. Investigating faculty perspectives and evaluating the impact of AI-specific academic integrity guidelines would help shape effective policies. Additionally, studies should assess the effects of AI-assisted learning on students' critical thinking, problem-solving, and knowledge retention to optimize AI integration in medical education while upholding academic integrity.

Beyond research, there is a clear need to integrate structured AI training into university curricula for medical students, interns, and healthcare professionals. Such training should not only cover the technical skills required to use AI tools effectively but also emphasize ethical considerations, academic integrity, and the professional boundaries of AI-assisted work. By embedding AI literacy and ethics into formal education, institutions can promote the use of AI for legitimate learning and professional development while reducing the risk of misuse for plagiarism or academic fraud. The development of structured guidelines and educational modules on AI ethics and best practices, implemented both at the Ministry level and within individual universities, could serve as a key step toward fostering responsible AI use in medical education and ensuring it supports, rather than undermines, professional development.

Conclusion

This study highlights the complex relationship between AI use and academic integrity among medical students. While AI tools are widely utilized for learning and assessment, perceptions regarding their ethical implications vary. The findings suggest a need for clearer institutional guidelines to define acceptable AI usage in academic settings. Given the potential risks of academic misconduct, integrating structured AI literacy programs into medical education could promote responsible AI use while maintaining academic integrity. Future research should further explore the evolving role of AI in education and its long-term impact on students' ethical decision-making and professional development.

Abbreviations

AI	Artificial intelligence.
UNESCO	United Nations Educational, Scientific and Cultural Organization.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

Study design: LL, IK; Data collection, analysis and interpretation: LL, KB, LP; Writing first draft of the manuscript: LL, LP; Revising the manuscript for content: LL, IK, KB, LP; Final approval of the manuscript: LL, IK, KB, LP.

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Data availability

Raw data collected during the study are available from the author Lesya Lymar on request (contact: lesyalymar@nmu.ua).

Declarations

Ethics approval and consent to participate

The Bogomolets National Medical University has a policy that the cross-sectional studies on healthy participants are exempt from the need to obtain approval of the ethics committee, i.e., such studies are eligible for a waiver. The Ethics Committee of the Bogomolets National Medical University waived the need for an ethics appraisal of the study protocol, as it was a cross-sectional study on healthy volunteers (official document to confirm the waiver was issued with a decision number 120/3 from December 27, 2024). The participants voluntarily participated in the survey upon being sent an invitation to participate, along with detailed information about the study, through Google Classroom. All participants provided written informed consent in the Google Form before starting to participate in the survey. Participants were allowed to withdraw from or stop the survey at any point before submitting it. The Google Form contained no previous information that could affect the answers. The study was carried out in compliance with the Helsinki declaration, and in line with all the applicable laws and regulations. To prevent unauthorized access, the first author (LL) stored the data on a password-protected server.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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